

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404203
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Jankovsky; Jiri et al.

Fibre producing machine with improved maintenance

Abstract

A machine for fiberizing mineral wool by free centrifugation, includes a frame on which is mounted at least one centrifugation wheel, the centrifugation wheel being connected to a transmission shaft designed to rotate it by transmitting the movement of rotation of an output shaft of a drive unit, wherein the machine further includes at least one intermediary transmission box connected at least by one input to the drive rod and by one output to the transmission shaft, the intermediary transmission box being arranged such as to transmit the movement of rotation of the drive shaft to the transmission shaft.

Inventors:	Jankovsky; Jiri (Veliny, CZ), Morin; Philippe (Hradec Kralove, CZ)
Applicant:	SAINT-GOBAIN ISOVER (Courbevoie, FR)
Family ID:	66542414
Assignee:	SAINT-GOBAIN ISOVER (Courbevoie, FR)
Appl. No.:	17/416241
Filed (or PCT Filed):	December 17, 2019
PCT No.:	PCT/FR2019/053103
PCT Pub. No.:	WO2020/128287
PCT Pub. Date:	June 25, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220055940 A1	Feb. 24, 2022

Foreign Application Priority Data

FR	1873401	Dec. 19, 2018
----	---------	---------------

Publication Classification

Int. Cl.: C03B37/05 (20060101); D01D5/08 (20060101); D01D5/18 (20060101)

U.S. Cl.:

CPC C03B37/055 (20130101); D01D5/08 (20130101); D01D5/18 (20130101);

Field of Classification Search

CPC: C03B (37/05-055); D01D (5/18)

USPC: 474/84; 474/86; 474/88; 474/148

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2398707	12/1945	Hawthorne	65/520	C03B 37/055
3191452	12/1964	Lipski	474/58	F16H 1/006
5131935	12/1991	Debouzie	65/456	C03B 37/055
5411444	12/1994	Nakamura	474/167	F16H 55/36

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2608498	12/2003	CN	N/A
439 385	12/1990	EP	N/A

OTHER PUBLICATIONS

International Search Report as issued in International Patent Application No. PCT/FR2019/053103, dated Mar. 13, 2020. cited by applicant

Primary Examiner: Snelting; Erin

Attorney, Agent or Firm: Pillsbury Winthrop Shaw Pittman LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. National Stage of PCT/FR2019/053103, filed Dec. 17, 2019, which in turn claims priority to French patent application number 1873401 filed Dec. 19, 2018. The content of these applications are incorporated herein by reference in their entireties.

(2) The present invention relates to the field of machines for manufacturing mineral wool.

PRIOR ART

(3) In order to manufacture mineral wool, it is known to use a machine **1** for fiberizing mineral wool by free centrifugation, as may be seen in FIGS. **1a** and **1b**. This machine for fiberizing mineral wool by free centrifugation conventionally comprises a series of spindles, each bearing a centrifugation wheel or rotor **2**. The spindles are arranged in parallel with one another in an

assembly placing the peripheral surfaces of the wheels close to one another.

(4) Hitherto, in such a fiberizing machine, the spindles are driven in rotation by a drive **3**. The bearings of the spindle are of the ball bearing or roller bearing type. They are lubricated by an oil mist and generally allow speeds of rotation of the shaft **4** of the order of 10000 revolutions per minute (10000 t/min).

(5) More precisely, the spindles are driven in rotation by a drive via the use of a connecting shaft **4**. This connecting shaft is used not only to form the link between the drive and the spindle carrying the rotor, but also to supply the binder used in the production of the mineral wool and even to supply and to circulate the cooling liquid for the spindles.

(6) In point of fact, currently, each shaft is in direct engagement with the drive either by screwing of the connecting shaft **4** to the drive shaft or by use of a belt or chain between the drive shaft and the shaft connected to the spindle. When a belt is used, the drive shaft and the connecting shaft are provided with a pulley that makes it possible to modify the gearing-down and to turn the connecting shaft more quickly.

(7) A single drive may be provided to turn all the rods.

(8) Thus, this direct engagement of the connecting shaft with the drive involves the drawback of maintenance being complex.

(9) In effect, such a configuration of direct engagement requires the maintenance operative to uncouple the rod from the drive and even to remove the pulleys and the belt in order to carry out maintenance or to make a repair. In point of fact, this operation is complex and time-consuming, and requires the apparatus to be re-set.

(10) Furthermore, this direct engagement also presents the drawback of limiting the possibilities of increasing the speed of rotation and thus of evolutions. In effect, the direct engagement of the rotors with the drives means that the maximum speed of the rotors is the maximum speed of the drives. Thus, the increase in the speed of rotation of the rotors may be achieved only by changing the drive and installing a drive that allows a higher speed of rotation. Such a change of drive is costly because it requires a more highly performing drive. It is also complex because it requires a change of drive for each rotor.

SUMMARY OF THE INVENTION

(11) The present invention thus proposes to provide a machine for fiberizing mineral wool by free centrifugation that resolves the prior-art drawbacks.

(12) To that end, the invention relates to a machine for fiberizing mineral wool by free centrifugation, comprising a frame on which is mounted at least one centrifugation wheel, said centrifugation wheel being connected to a transmission shaft designed to rotate it by transmitting the movement of rotation of an output shaft of a drive unit, characterized in that said machine further comprises at least one intermediary transmission box connected at least by one input to the drive rod and by at least one output to a transmission shaft, said intermediary transmission box being arranged such as to transmit the movement of rotation of the drive shaft to the transmission shaft.

(13) The present invention advantageously makes it possible to have a machine for fiberizing rockwool for which maintenance is simplified. In effect, the presence of an intermediary transmission box advantageously makes it possible to not connect the drive directly to the shafts on which the rotors are arranged. Thus, a change of belt or of pinions takes place by removing the intermediary transmission box without dismantling the transmission shafts.

(14) According to one example, the machine comprises a plurality of centrifugation wheels, each centrifugation wheel being connected to a transmission shaft and comprising an intermediary transmission box for each transmission shaft.

(15) According to one example, each transmission box comprises a chassis in which a rod is mounted, said rod comprising coupling means for coupling to the transmission shaft and to the drive.

- (16) According to one example, each transmission box comprises a chassis in which a first rod and a second rod are arranged and extend in parallel in the same direction, the first rod and the second rod are mechanically coupled to coupling elements, the first rod and the second rod each comprise coupling means in order to be connected, respectively, to the transmission shaft and to the drive unit.
- (17) According to one example, a single transmission box is arranged, and this transmission box is arranged such that all the transmission shafts are connected thereto.
- (18) According to one example, said transmission box comprises a chassis in which a plurality of rods extending in parallel in the same direction is arranged, each rod comprising coupling means in order for each one to be connected to a transmission shaft and to the drive unit.
- (19) According to one example, said transmission box comprises a chassis in which, for each transmission shaft, a first rod and a second rod are arranged and extend in parallel in the same direction, the first rod and the second rod are mechanically coupled to coupling elements, the first rod and the second rod each comprise coupling means in order to be connected, respectively, to the transmission shaft and to the drive unit.
- (20) According to one example, the coupling means and/or the coupling elements comprise at least two pulleys each fixed to a rod and connected in pairs by a belt.
- (21) According to one example, the coupling means and/or the coupling elements comprise at least two pinions for coupling two rods together.
- (22) According to one example, the coupling means and/or the coupling elements comprise at least two pinions meshing together for coupling two rods together.
- (23) According to one example, the coupling means and/or the coupling elements are arranged such as to allow an output rod to have a speed of rotation that is different from that of an input rod, preferably such as to have a higher speed of rotation.
-

Description

DESCRIPTION OF THE FIGURES

- (1) Other particular features and advantages will become clearly apparent from the following description made thereof by way of indication and in a completely non-limiting manner, with reference to the appended drawings in which:
- (2) FIGS. **1a** and **1b** are schematic representations of a fiberizing machine according to the prior art;
- (3) FIGS. **2**, **3a**, **3b**, and **3c** are schematic representations of a fiberizing machine according to the invention;
- (4) FIGS. **4** to **8** are schematic representations of different implementations of the first embodiment of the invention;
- (5) FIGS. **9** to **14** are schematic representations of the second embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

- (6) FIGS. **2** and **3a** show a machine **100** for fiberizing mineral wool by free centrifugation according to the invention. Such a machine **100** for fiberizing mineral wool by free centrifugation comprises a frame **102** on which the various elements are mounted.
- (7) On the frame **102**, a plurality of spindles or centrifugation wheels or rotors **104**, at least 3 in number, preferably three or four or more, are arranged in cascade and are capable of turning about substantially horizontal axes, two successive wheels **104** on the path of the molten material turning in the opposite direction. The first centrifugation wheel **104** is supplied with molten material by a channel and essentially serves to accelerate the material, which is returned toward the second centrifugation wheel **104** and so on and so forth as far as the last centrifugation wheel, the flow of material diminishing at each centrifugation wheel **104** owing to the quantity of formed fibers.

(8) Such a machine **100** generally further comprises means for generating an air current (not shown) at the periphery of the centrifugation wheels with the aim of assisting the formation of the fibers by means of a drawing effect and of taking up the fiberized material, separating it from the non-fiberized material (pellets). This latter is undesirable as it contributes to making the final product heavier and particularly unpleasant to the touch. The air current also has the function of conveying the fiberized material toward a receiving member, for example an endless belt provided with suction compartments, which conveys the fibers toward the downstream processing devices of the line, such as a cross-lapping device, a binder polymerization oven, etc.

(9) The air current is usually introduced in a direction substantially parallel to the axes of rotation of the centrifugation wheels **104** and thus entrains the fibers in a direction perpendicular to their direction of formation. The quantity of fibers manufactured is dependent on the rate of flow of material poured onto the centrifugation wheels **104** and the fiberizing efficiency of said centrifugation wheels **104**.

(10) The centrifugation wheels **104** are connected to a drive unit M in order to be rotated. Each centrifugation wheel **104** is connected to a transmission shaft **110** allowing connection to the drive unit M. In FIG. 3c, the drive unit M comprises, in a first case, a single drive m associated with a gear unit N to make it possible to have a plurality of outputs m' and thus to entrain a plurality of centrifugation wheels **104** in rotation. In a second case, shown in FIG. 3b, the drive unit M comprises a plurality of independent drives m each having an output m'.

(11) The transmission shafts **110** connecting the centrifugation wheels **104** to the drive unit M may be solid or hollow. Where a transmission shaft **110** is hollow, it may comprise at least one conduit allowing the cooling of the wheels **104** and/or one conduit for supplying the binder to the centrifugation wheels **104** in order to manufacture the rock wool.

(12) Ingeniously, according to the invention, the transmission shafts **110** are not meshed directly with the drive unit M. In effect, the machine **100** according to the present invention further comprises intermediary transmission means **200** arranged between the drive unit M and each transmission shaft **110**.

(13) In a first embodiment that can be seen in FIG. 4, these intermediary transmission means **200** comprise a plurality of intermediary transmission boxes **201**, each intermediary transmission box **201** being connected to a transmission shaft **110**. It will be understood that, where there is a single drive m, the intermediary transmission boxes **201** are all connected to the outputs m' of said drive m, whereas in the case of a fiberizing machine **100** having a plurality of drives each intermediary transmission box **201** is connected to the output m' of a drive m.

(14) In a first implementation that can be seen in FIGS. 5a and 5b, each intermediary transmission box **201** comprises a chassis **201a** in which a rod **202** comprising a first end and a second end is arranged. This chassis **201a** may be faired (FIG. 5a) or unfaired (FIG. 5b). These ends are arranged such as to lie outside the faired or unfaired chassis. At its ends, the rod **202** comprises coupling means **203** for its coupling to the transmission shaft **110** and to the drive unit M.

(15) One advantage of this intermediary transmission box **201** according to the invention is that it allows easy assembly and dismantling by means of simple action on the coupling means **203** in order to uncouple said intermediary transmission box **201** from the drive unit M and from the transmission shaft **110** to which it is coupled. Thus, in the event of replacement or maintenance, it is easy to gain access to this intermediary transmission box **201**. Similarly, in the event of maintenance to be carried out on the centrifugation wheels **104**, it is easy to dismantle only the centrifugation wheels **104** and the transmission shafts **110** while keeping the intermediary transmission box **201** connected to the drive unit M.

(16) In a second implementation that can be seen in FIG. 6, each transmission box **201** comprises a chassis **201a** in which two rods **202a**, **202b** are mounted. The rods **202a**, **202b** are arranged such as to extend in parallel in the same direction. The first rod **202a** and the second rod **202b** are arranged such as to be coupled mechanically. This coupling is achieved by coupling elements **206**. The first

rod **202a** is arranged such as to be coupled to the drive m while the second rod **202b** is arranged such as to be coupled to the centrifugation wheel **104**.

(17) These coupling elements **206** comprise a belt system comprising a pulley on the rod **202a** and a pulley on the rod **202b**, a belt connecting the two pulleys.

(18) Alternately, these coupling elements **206** comprise a chain system comprising a toothed wheel on the rod **202a** and a toothed wheel on the rod **202b**, a chain connecting the two toothed wheels.

(19) Alternately, these coupling elements **206** comprise a gearing system comprising a pinion on the rod **202a** and a pinion on the rod **202b**. The pinions are meshed with one another directly or indirectly via other pinions.

(20) The first rod **202a** and the second rod **202b** also each comprise coupling means **203** such as to be connected, respectively, to the transmission shaft **110** and to the drive unit M.

(21) In a third implementation that can be seen in FIG. 7, each transmission box **201** comprises three rods **202a**, **202b**, **202c**. The rods **202a**, **202b**, **202c** are arranged such as to extend in parallel in the same direction. The first rod **202a** is arranged such as to be connected mechanically to the second rod **202b** and the third rod **202c** by means of coupling elements **206**. Coupling means **203** are arranged to connect the transmission box **201** to the transmission shaft **110** and to the drive unit M. The second rod **202b** is coupled to the drive unit M and the third rod **202c** is coupled to the transmission shaft **110**.

(22) It is possible for the coupling elements **206** to allow the coupling of the first rod **202a** to the second rod **202b** to be different from the coupling elements **206** allow the coupling of the first rod **202a** to the third rod **202c**.

(23) The advantage of the second implementation and of the third implementation is to allow the presence of standard coupling means **203** while permitting an increase in the speed of rotation.

(24) In effect, the presence of the coupling elements **206** allows the presence of a gearing-down between the rods **202a**, **202b**, **202c** in order to increase the speed of rotation of the centrifugation wheel **104** relative to the speed of rotation of the drive.

(25) This possibility enables gearing-down to be dispensed with at the coupling means **203**. The latter may thus be standard, i.e. identical for each intermediary transmission box **201** of one and the same fiberizing machine **100**.

(26) As a result, not only is it possible to retain an existing drive m in order to obtain a higher speed of rotation than it is able to provide but furthermore it is possible to invert the intermediate transmission boxes **201** between the different drives m according to requirements.

(27) For these different implementations of the first embodiment, the coupling means **203** may comprise a belt system comprising a pulley **2031** on the rod **202** and a pulley **2031** on the transmission shaft **110** or on an output m' of the drive unit. A belt **2032** connects the two pulleys **2031** as may be seen in FIG. 8a.

(28) Alternately, these coupling means **203** may comprise a chain system comprising a toothed wheel **2033** on the rod **202** and a toothed wheel **2033** on the transmission shaft **110** or on an output m' of the drive unit. A chain or notched belt **2034** connects the two toothed wheels as may be seen in FIG. 8b.

(29) Alternately, these coupling means **203** may comprise a gearing system comprising a pinion **2035** on the rod **202** and a pinion **2035** on the transmission shaft **110** or on an output m' of the drive unit M. The pinions **2035** are meshed with one another directly or indirectly via other pinions **2035** as may be seen in FIG. 8c.

(30) Alternately, the shaft **110** may be in direct engagement with the rod **202** of the intermediary transmission box **200** via screws or teeth extending in the same direction as the rod. Naturally, in the case of the first implementation, the coupling means **203** are, optionally, arranged such as to allow a gearing-down, increasing the speed of the centrifugation wheel **104** relative to the speed of the drive m.

(31) This intermediary transmission box **201** also offers the advantage of enabling the

centrifugation wheel **104** to not lie in the axis of the drive rod m' . It is thus possible to position the centrifugation wheels **104** as desired.

(32) In a second embodiment that can be seen in FIGS. **9** to **11**, the fiberizing machine **100** according to the present invention comprises intermediary transmission means **200** arranged such that all the transmission shafts **110** are connected thereto. To that end, the intermediary transmission means **200** comprise a single intermediary transmission box **210**.

(33) In a first implementation of this second embodiment, this single intermediary transmission box **210** comprises a chassis **210a** in which a plurality of rods **212** is arranged. The number of rods **212** is equal to the number of wheels **104**. Thus, each rod **212** comprises coupling means **203** in order to be coupled to the drive unit M and to the centrifugation wheels **104** as may be seen in FIGS. **10**, **11**, and **14**.

(34) In a second implementation to this second embodiment that can be seen in FIG. **12**, each rod **212** is replaced by two rods **212a**, **212b** arranged such as to extend in parallel in the same direction. The first rod **212a** and the second rod **212b** are arranged such as to be coupled mechanically by coupling elements **206**. The first rod **212a** is arranged such as to be coupled to the drive m while the second rod **212b** is arranged such as to be coupled to the centrifugation wheel **104**.

(35) In a third implementation to this second embodiment that can be seen in FIG. **13**, each transmission box **210** comprises three rods **212a**, **212b**, **212c**. The rods **212a**, **212b**, **212c** are arranged such as to extend in parallel in the same direction. The first rod **212a** is arranged such as to be mechanically with the second rod **212b** and the third rod **212c** by means of coupling elements **206**. The second rod **212b** is coupled to the drive unit M by coupling means **203**. The third rod **212c** is coupled to the transmission shaft **110** by coupling means **203**. In the case of FIG. **13**, the coupling means **203** coupling the transmission shaft **110** to the transmission box **210** are simply bolting of said shaft **110** to the rod **212c** of the transmission box **210**.

(36) For these different implementations of the second embodiment, the coupling means **203** may comprise a belt system comprising a pulley on the rod **202** and a pulley on the transmission shaft **110** or on an output m' of the drive unit. A belt connects the two pulleys.

(37) Alternately, these coupling means **203** may comprise a chain system comprising a toothed wheel on the rod **202** and a toothed wheel on the transmission shaft **110** or on an output m' of the drive unit. A chain connects the two toothed wheels.

(38) Alternately, these coupling means **203** may comprise a gearing system comprising a pinion on the rod **202** and a pinion on the transmission shaft **110** or on an output m' of the drive unit M . The pinions are meshed with one another directly or indirectly via other pinions.

(39) Alternately, the shaft **110** may be in direct engagement with the rod of the intermediary transmission box **200** via screws or teeth extending in the same direction as the rod.

(40) Naturally, in a variant of the different implementations, the coupling means **203** or the coupling elements **206** are arranged such as to make it possible to modify the speed between an input rod, connected directly or indirectly to the drive unit M , and an output rod connected directly or indirectly to the centrifugation wheel **104**. This variation in the speed of rotation is obtained by modifying the gearing or the pulleys or the toothed wheels forming coupling elements **206** and/or the coupling means **203** such as to obtain a gearing-down.

(41) In the case of coupling means **203** and/or coupling elements **206** comprising pulleys connected by a belt, the pulley of the output rod will be of smaller diameter than that of the input rod.

(42) In the case of coupling means **203** and/or coupling elements **206** comprising toothed wheels connected by a chain, the toothed wheel of the output rod will be of smaller diameter and will have fewer teeth than that of the input rod.

(43) In the case of coupling means **203** and/or of coupling elements **206** comprising a gearing comprising at least two pinions meshed with one another, the pinion of the output rod will be of smaller diameter and will have fewer teeth than that of the input rod.

(44) This variant advantageously makes it possible to be able to increase the speed of rotation of

the rotors/centrifugation wheels **104** without the need for a more highly performing drive unit M. It becomes easy to modify the coupling means **203** or the coupling elements **206** in order to adapt to the desired speed of rotation.

(45) Furthermore, having intermediary transmission boxes **210** with identical coupling means **203** advantageously makes it possible to standardize these components, which tend to wear. As a result, this standardization affords an advantage in terms of maintenance as there is only a single model of intermediary transmission box **210**.

(46) In a variant of the embodiments, the transmission shaft **110** is composed of two half-shafts. These half-shafts are connected to one another by coupling means that may be a belt associated with two pulleys or a chain associated with two toothed wheels or two pinions meshing with one another. A first half-shaft is connected to the intermediary transmission box while the second half-shaft is connected to the centrifugation wheel **104** used for fiberizing.

(47) This variant advantageously makes it possible to have standard intermediary transmission boxes **201**, **210** and to implement an increase in the speed of rotation of the associated rotor. In effect, the coupling elements are designed such that the second half-shaft connected to the centrifugation wheel **104** has a speed of rotation greater than the speed of rotation of the first half-shaft connected to the intermediary transmission box **201**, **210**. This is possible by configuring the pulleys or the toothed wheels or the pinions such that the pulley or the toothed wheel or the pinion installed on the second half-shaft is smaller than the pulley or the toothed wheel or the pinion installed on the first half-shaft. In the case of the toothed wheel or of the pinion, this reduction in size is accompanied by a reduction in the number of teeth.

(48) Naturally, the present invention is not limited to the example illustrated but may be the subject of diverse variants and modifications that will be apparent to a person skilled in the art.

(49) Thus, it is possible to design the intermediary transmission boxes **201**, **210** such that the coupling means **203** and/or the coupling elements **206** are different in nature.

Claims

1. A machine for fiberizing mineral wool by free centrifugation, the machine extending along a longitudinal direction and comprising: a drive unit positioned at a first location along the longitudinal direction and including an output rod that extends along the longitudinal direction; a frame positioned at a second location along the longitudinal direction and on which is mounted at least one centrifugation wheel, said at least one centrifugation wheel being connected to a transmission shaft extending parallel to said longitudinal direction and adapted to rotate the at least one centrifugation wheel by transmitting a movement of rotation of the output rod of the drive unit, and at least one intermediary transmission box connected at least by one input to the output rod of the drive unit and by at least one output to the transmission shaft, said at least one intermediary transmission box being arranged at an intermediate location along the longitudinal direction that is between the first location and the second location and between the output rod and the transmission shaft such as to transmit the movement of rotation of the output rod of the drive unit to the transmission shaft, wherein the output rod is connected to the at least one intermediary transmission box using a first belt and at least one first pulley, wherein the belt is coupled to a separate rod arranged in the at least one intermediary transmission box, wherein the separate rod is coupled to the transmission shaft via a second belt and a second pulley, and wherein the second belt and the second pulley are arranged between the at least one intermediary transmission box and the at least one centrifugation wheel.

2. The fiberizing machine as claimed in claim 1, comprising a plurality of centrifugation wheels, each centrifugation wheel being connected to a transmission shaft and comprising an intermediary transmission box for each transmission shaft.

3. The fiberizing machine as claimed in claim 2, wherein each transmission box comprises a

chassis in which a rod is mounted, said rod comprising coupling means for coupling to the transmission shaft and to the drive unit.

4. The fiberizing machine as claimed in claim 3, wherein the coupling means comprise at least two pulleys each fixed to a rod and connected in pairs by a belt.
 5. The fiberizing machine as claimed in claim 4, wherein the coupling means are arranged such as to allow the output rod to have a speed of rotation that is different from that of the transmission shaft.
 6. The fiberizing machine as claimed in claim 5, wherein the transmission shaft has a speed of rotation higher than that of the output rod.
 7. The fiberizing machine as claimed in claim 3, wherein the coupling means comprise at least two pinions for coupling two rods together.
 8. The fiberizing machine as claimed in claim 3, wherein the coupling means comprise at least two pinions meshing together for coupling two rods together.
 9. The fiberizing machine as claimed in claim 2, wherein each transmission box comprises a chassis in which a first rod and a second rod are arranged and extend in parallel in the same direction, the first rod and the second rod are mechanically coupled to coupling elements, the first rod and the second rod each comprise coupling means in order to be connected, respectively, to the transmission shaft and to the drive unit.
 10. The fiberizing machine as claimed in claim 1, comprising a plurality of transmission shafts and a plurality of centrifugation wheels that are each connected to one of the plurality of transmission shafts, wherein the at least one intermediary transmission box is a single transmission box, and said single transmission box is arranged such that all of the plurality of transmission shafts are connected thereto.
 11. The fiberizing machine as claimed in claim 10, wherein said single transmission box comprises a chassis in which a plurality of rods extending in parallel in a same direction is arranged, each rod comprising coupling means in order for each one to be connected to one of the plurality of the transmission shafts and to the drive unit.
 12. The fiberizing machine as claimed in claim 10, wherein said single transmission box comprises a chassis in which, for each transmission shaft, a first rod and a second rod are arranged and extend in parallel in a same direction, the first rod and the second rod are mechanically coupled to coupling elements, the first rod and the second rod each comprise coupling means in order to be connected, respectively, to the transmission shaft and to the drive unit.
 13. The fiberizing machine as claimed in claim 1, wherein the at least one intermediary transmission box comprises a chassis in which a rod is mounted, said rod being coupled to the output rod of the drive unit and to the transmission shaft.
 14. The fiberizing machine as claimed in claim 1, wherein the at least one centrifugation wheel comprises a first and a second centrifugation wheel connected to, respectively, a first and a second transmission shaft.
 15. The fiberizing machine as claimed in claim 1, wherein the first belt and the at least one first pulley are arranged between the at least one intermediary transmission box and the drive unit.
 16. The fiberizing machine as claimed in claim 1, wherein the separate rod is parallel to the longitudinal direction.
-